cleared misbranded. Another IndoFil product met the same fate in February 2025. "There's a silent mafia at work. Unknown brands and counterfeit products are flooding the market. The traders are let off with negligible fines, their licences aren't cancelled, and there's no real deterrent," alleges Mushtaq Ahmad Malik, president of the Zamindar Association Shopian, while backing the demand for crop insurance as a much-needed safety net.

Delays in testing compound the problem. Of the 103 pesticide molecules in use, only 49 can be tested in Srinagar. The rest are sent to the Central Insecticide Laboratory in Faridabad, delaying red flags and allowing further damage to orchards. But Mujtaba Yahya, deputy director, law enforcement, claims the "farmers

KASHMIR ACCOUNTS FOR OVER 75 PER CENT OF INDIA'S APPLE PRODUCTION. BETTER CHECKS ARE NEEDED TO SECURE THIS VITAL SECTOR

are exaggerating the issue". "We have round-the-clock checks," he adds, though admitting that "modern labs in every district would bolster our efforts". The directorate lacks resources—no separate budget and just nine inspectors covering the entire Valley.

Efforts are underway. "Two new pesticide testing labs have started this year, in Kulgam and Baramulla. We're working to control the diseases and protect farmers," says Zahoor Ahmad Bhat, director of the horticulture department in Kashmir. But for growers like Gulzar Ahmad Bhat, relief feels distant. As Kashmir's apple industry grapples with rising risks and regulatory lapses, only swift reforms and stronger safeguards will ensure that one of the region's most vital livelihoods survives and thrives. ■

SCIENCE

CRACKING INDIA'S GENETIC CODE

By Ajay Sukumaran

India's largest-ever genomic study has identified genetic variants unique to the country, marking a crucial step in closing a longstanding gap in global genomics research. The preliminary findings, published last week in *Nature Genetics*, come from the GenomeIndia project—a nationwide effort to map the country's vast genetic diversity.

Launched in January 2020, GenomeIndia brought together over 100 researchers from 20 institutions to conduct whole genome sequencing—a technique by which the sequence of the building blocks of DNA, known as nucleotides, is decoded—across a broad swathe of India's population. The study analysed DNA from 10,074 individuals, drawn from a total of 20,000 blood samples collected from 83 population groups—both tribal and non-tribal.

Despite being home to more than 4,600 distinct ethnic groups, India's representation in global genomics databases has long been lacking. The project aims to generate vital data on the genetic makeup of the Indian population, with implications for better understanding of health and disease patterns unique to the region.

A human genome has 3.2 billion base pairs, or sequences formed by the four nucleotides [adenine (A), cytosine (C), guanine (G) and thymine (T)]. These pairs are organised into long molecules of DNA called chromosomes of which there are 23 pairs—22 of these are called autosomes and one pair comprises the sex chromosomes which decide the individual's sex. Typically, two individuals can differ by 0.1 to 0.4 per cent of their genome, which means there will be differences in a few million positions in their DNA code.

The GenomeIndia study found 180 million genetic variants from the whole genome sequencing of 9,772 individuals. Of these, around 130 million variants were in the autosomes, or non-sex chromosomes. As much as 65 per cent of variants were ultra-rare. "The identified genetic variants decode extensive genetic diversity that has been hitherto uncaptured in the Indian population," the paper said.

"This is just the beginning. The GenomeIndia team is now performing in-depth analysis to find out the meaning of these genetic variations," says Dr Kumarasamy Thangaraj, joint national coordinator of GenomeIndia and CSIR Bhatnagar Fellow at the CSIR-Centre for Cellular & Molecular Biology at Hyderabad. Some variants, he explains, may be associated with diseases; others with therapeutic/ adverse response to drugs, susceptibility/ resistance to infectious disease, or adaption. "Once the information from the genome sequence of 9,772 individuals is extracted, we will have very comprehensive information about the Indian population, its history as well as medical implications."

There are two promising areas of application in healthcare: personalised medicine and